# Huawei Access Node HSR-ACC-74 QoS Congestion Event Resolution SOP

**Node:** HSR-ACC-74 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_3153

## Description

The Huawei Access Node HSR-ACC-74, operating at the HSR Layout Access site, has experienced a Major failure event (Alarm Code: ALARM_3153) attributed to a QoS / Congestion Event failure domain. This failure domain is primarily a result of the occurrence of microbursts and interface tail drops, which in turn precipitate QoS queue starvation. This phenomenon can be attributed to the interaction of multiple factors, including the dynamic nature of QoS mapping in Huawei's Access Node architecture, the inherent limitations of the QoS queueing mechanisms (e.g., WFQ and PQ), and the impact of microbursts and tail drops on the access node's forwarding table and queue management systems. Furthermore, the specific role of this node as an access device, with its focus on delivering traffic from end-user devices to the network core, exacerbates the sensitivity of this equipment to QoS-related failures.

## Pre-requisite Checks

- Confirm the site location (HSR Layout Access) and verify the node ID (HSR-ACC-74)
- Verify the CLI dialect being used is Huawei-specific
- Check the redundant link status to ensure it is active and functioning correctly
- Dispatch remote hands to the site location (HSR Layout Access) if necessary
- Verify the power supply unit (PSU) and fan tray are operational

## Potential Root Causes

| Cause | Probability |
|---|---|
| Microburst-induced QoS queue starvation | High |
| Interface tail drops due to congestion | Medium |
| QoS mapping conflicts and misconfigurations | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display queue statistics, Expected Output: Queue statistics for the QoS queues, Description: This command is used to isolate the QoS queue starvation issue by displaying the queue statistics for the QoS queues. It helps identify which queues are experiencing congestion and starvation, thereby isolating the root cause of the issue.

Command: display interface statistics, Expected Output: Interface statistics for the access node interfaces, Description: This command is used to isolate the interface tail drops due to congestion by
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```
displaying the interface statistics for the access node interfaces. It helps identify which interfaces are experiencing congestion and tail drops, thereby isolating the root cause of the issue.

Command: display qos map, Expected Output: QoS mapping configuration for the access node, Description: This command is used to isolate the QoS mapping conflicts and misconfigurations by displaying the QoS mapping configuration for the access node. It helps identify any conflicts and misconfigurations in the QoS mapping, thereby isolating the root cause of the issue.
```

## Resolution Steps

Step 1: Step: Clear the QoS queues, Command: clear qos queue, Description: This step involves clearing the QoS queues to remove any existing congestion and starvation issues. The command "clear qos queue" is used to clear the QoS queues and restore normal operation.

Step 2: Step: Adjust the QoS mapping configuration, Command: set qos map, Description: This step involves adjusting the QoS mapping configuration to resolve any conflicts and misconfigurations identified during the diagnostic phase. The command "set qos map" is used to adjust the QoS mapping configuration and resolve the issue.

Step 3: Step: Verify the interface settings, Command: display interface, Description: This step involves verifying the interface settings to ensure they are configured correctly and not contributing to the congestion and tail drops. The command "display interface" is used to verify the interface settings and identify any issues.

## Post-Check Verification

```
Command: display queue statistics, Expected Output: Queue statistics for the QoS queues (normal), Description: This command is used to verify that the QoS queue starvation issue has been resolved by displaying the queue statistics for the QoS queues. It helps confirm that the QoS queues are operating normally and not experiencing congestion and starvation.

Command: display interface statistics, Expected Output: Interface statistics for the access node interfaces (normal), Description: This command is used to verify that the interface tail drops due to congestion have been resolved by displaying the interface statistics for the access node interfaces. It helps confirm that the interfaces are operating normally and not experiencing congestion and tail drops.

Command: display qos map, Expected Output: QoS mapping configuration for the access node (corrected), Description: This command is used to verify that the QoS mapping conflicts and misconfigurations have been resolved by displaying the QoS mapping configuration for the access node. It helps confirm that the QoS mapping is correct and not contributing to the issue.
```

## Roll-back Procedures

- Step: Restore the original QoS mapping configuration, Command: restore qos map, Description: This step involves restoring the original QoS mapping configuration in case the fix fails. The command "restore qos map" is used to restore the original QoS mapping configuration and return the system to its previous state.

- Step: Clear the QoS queues, Command: clear qos queue, Description: This step involves clearing the QoS queues in case the fix fails. The command "clear qos queue" is used to clear the QoS queues and restore normal operation.